

MANAR: An Evidence-Scoped Software Testbed for Human-Supervised Aerial Search and Rescue

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Abstract—Search-and-rescue unmanned aircraft must automate repetitive search tasks without allowing uncertain perception to make high-consequence decisions. This paper presents MANAR (Multi-sensor Autonomous Navigation and Aerial Rescue), an evidence-scoped software testbed that separates implemented source, measured software behavior, and planned flight capabilities. The current artifact contains a C++ control core with Haversine navigation, nearest-neighbor sector ordering, operator-preserved manual ordering, plan locking, and Return-to-Launch semantics, together with ONNX Runtime detector harnesses. A fixed-seed study of 500 synthetic sector layouts finds that greedy tours are 4.6%–11.1% longer on average than tours refined by 2-opt; 2-opt is treated as a local-search reference, not a global optimum. A diagnostic zero-shot comparison on 26 ten-second clips (6,240 frames) reports mean model-session latency of 29.25 ± 0.85 ms for YOLO11n and 50.53 ± 3.15 ms for D-FINE-N across clip means. At least one person event occurs in 12 of 15 positive-designated clips for YOLO11n and 6 of 15 for D-FINE-N; neither produces an event in 11 negative-designated clips. Because the corpus lacks bounding-box ground truth, these counts are not precision, recall, or average precision. No physical flight, calibrated multisensor fusion, or operational rescue performance is claimed.

Index Terms—Search and rescue, supervised autonomy, route planning, aerial perception, software testbed, reproducibility.

I. INTRODUCTION

SEARCH-AND-RESCUE (SAR) operations in wilderness, desert, and mountainous terrain are time-critical and cognitively demanding. Unmanned aerial vehicles (UAVs) can extend visual coverage and reduce responder exposure, but field experience shows that autonomy must be coordinated with human judgment rather than treated as a substitute for it [1]–[3]. A vehicle may be able to generate a sweep, reorder sectors, or produce a person detection while still lacking the evidence required to declare a rescue target or command a safety-critical maneuver.

MANAR (Multi-sensor Autonomous Navigation and Aerial Rescue) is developed around this distinction. The project combines a deterministic C++ control prototype, route-ordering logic, and a zero-shot aerial-perception harness, while reserving target confirmation and guidance authorization for a human operator. The research question in this paper is therefore not whether MANAR is already an autonomous rescue aircraft. It is: *what credible pre-flight evidence can the present software artifact provide, and where must its claims stop before simulation, hardware integration, annotation, and flight trials?*

This framing matters because a software repository can easily mix three different types of statement. Source code can establish

TABLE I
EVIDENCE SCOPE USED FOR MANAR CLAIMS

Status	Meaning in this paper
Implemented	Source and configuration artifacts are present in the repository; this does not by itself prove flight readiness.
Measured	A stated script, input corpus, aggregation rule, and result artifact support the observation.
Planned	A design target or selected architecture has not yet been demonstrated end to end.

that a mechanism is present; an experiment can establish behavior under a defined protocol; and a design document can specify a future capability. None is interchangeable with another. Table I defines the vocabulary used throughout this paper.

A. Related Work and Research Gap

Human-supervised SAR robotics: Wilderness SAR studies emphasize operator workload, search-pattern support, and human interpretation of ambiguous imagery [1]. Integrated urban SAR platforms likewise separate flight stabilization from higher-level mission and perception software so that failures remain contained [4]. MANAR adopts the same authority boundary but focuses on making the maturity of each software claim explicit.

Coverage and route planning: UAV coverage planning commonly uses parallel-track or Boustrophedon sweeps, with orientation and energy constraints affecting mission cost [5]–[7]. Ordering disconnected sectors is related to the metric Traveling Salesperson Problem (TSP) [8]. MANAR uses nearest-neighbor construction because its behavior is deterministic and easy to inspect, but evaluates it only against a specified 2-opt local-search reference rather than calling that reference globally optimal.

Aerial and multispectral perception: Aerial detection is dominated by small targets, oblique viewpoints, motion, and clutter [9], [10]. Slicing can preserve small targets that would otherwise disappear during resizing [11]. Thermal and multispectral pedestrian research demonstrates the value of complementary channels while also requiring calibrated sensors, synchronized data, and valid annotation [12], [13]. The present MANAR clip corpus uses RGB-, infrared-, and thermal-labeled conditions, but the repository does not yet document calibrated physical sensor provenance; this paper therefore treats it as a diagnostic multi-condition corpus rather than a validated multispectral benchmark.

Real-time edge inference: YOLO-style dense detectors and real-time transformer detectors offer different accuracy–latency

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tradeoffs [14]–[16]. Deployment also depends on preprocessing, graph export, postprocessing, and target hardware. A two-model zero-shot comparison cannot isolate a universal architectural cause, so MANAR reports an observed difference and a hypothesis to test rather than a general verdict on convolutional and transformer detectors.

The gap addressed here is consequently narrower than a new detector or flight controller: a reproducible, evidence-scoped account of what a pre-flight SAR software testbed presently establishes, including negative results and threats to validity.

B. Research Questions and Contributions

The paper evaluates three research questions:

- **RQ1—Route quality:** How much longer is deterministic nearest-neighbor ordering than a 2-opt-refined tour on fixed-seed synthetic sector layouts?
- **RQ2—Diagnostic perception behavior:** Under a fixed zero-shot protocol, how do YOLO11n and D-FINE-N differ in model-session latency, raw detection-event count, and positive-scenario coverage?
- **RQ3—Evidence boundary:** Which MANAR mechanisms are represented in source, which observations are measured, and which operational capabilities remain planned?

The contributions are:

- 1) an explicit evidence taxonomy applied to a human-supervised SAR software architecture, preventing source presence, diagnostic measurement, and future design from being conflated;
- 2) a fixed-seed, 500-instance reference evaluation of nearest-neighbor sector ordering against best-improvement 2-opt, with the script and machine-readable results included in docs/paper/reproducibility/; and
- 3) a corrected diagnostic report for 26 local clips (6,240 frames) that uses only metrics supported by the benchmark artifact and explicitly identifies missing ground truth, timing asymmetry, and external-validity limits.

No physical airframe performance, flight endurance, calibrated multisensor fusion, bounding-box accuracy, or operational rescue success is claimed.

II. SYSTEM REQUIREMENTS

MANAR uses the following operational and safety requirements to drive software design. Values in this section are *design targets or simulated configuration limits*, not measured airframe capabilities.

A. Operational Envelope and Performance Targets

The system targets an operational flight envelope tailored for rugged wilderness and desert environments:

- **Search Endurance Target (SYS-G-02):** At least 1.0 hour of continuous active search under a nominal payload and representative sensor duty cycle; physical endurance has not been measured.
- **Configured Altitude Limit (SYS-G-01):** The software test configuration clamps commanded altitude at 2000.0 m.

TABLE II
SUPERVISED-AUTONOMY RESPONSIBILITY ALLOCATION

Layer	Core Responsibilities
Human Operator	Mission definition, geofencing, takeoff approval, real-time RTL/waypoint override, alert review, final probable-rescue classification, and guidance authorization.
Deterministic Core	Implemented prototype navigation, route ordering, configuration clamps, plan locking, and RTL state changes; future autopilot integration remains planned.
Machine Learning	Implemented raw visual candidate detections; planned acoustic/radar classification and temporal fusion. Machine-learning output is advisory and does not directly command actuators.

This is not evidence of climb, hover, regulatory, or propulsion capability at that altitude.

- **Speed Presets and Limits:** Four discrete velocity presets (v_Q, v_A, v_I, v_H) bounded by software velocity clamp $v_{\max}$ and altitude ceiling $H_{\max}$. Takeoff climb target is specified by H_{launch} .

B. Terrain-Adaptive Search Strategy

Search path generation adapts to local terrain geometry:

- **Open / Flat Terrain (implemented source):** Parallel-track rectangular sweeps parameterized by sector width W , height H , and row spacing Δy .
- **Mountainous / Steep Terrain (planned):** Terrain-aware contour following with a verified digital elevation model and sensor standoff constraint.

C. Supervised-Autonomy Responsibility Allocation

Table II delineates the functional boundaries between the human operator, deterministic control logic, and machine learning pipelines.

D. Deterministic Contingency and Failsafe Rules

The target flight system defines deterministic contingency responses. Only software mechanisms explicitly identified elsewhere as implemented should be read as present in the current artifact:

- **Command-Link Loss (SYS-R-01, planned integration):** Hold, timeout, and RTL behavior requires autopilot and communication-link simulation before verification.
- **Energy Failsafe (SYS-R-02, simulated source):** Threshold-driven warning, RTL, and landing states can be exercised with the discrete battery test variable; return-energy sufficiency is not physically modeled.
- **GNSS Degradation (planned):** Attitude hold and local clearance sensing depend on future autopilot and sensor integration.
- **Environmental Limits (planned):** Wind, icing, and propulsion abort logic requires measured sensors and flight testing.

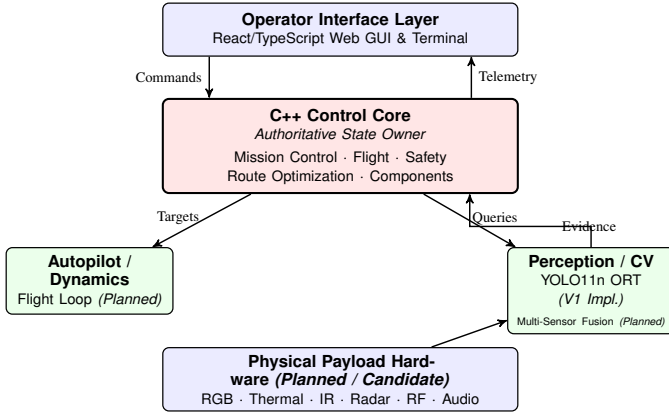


Fig. 1. MANAR system architecture. The C++ Control Core is the authoritative state owner governing navigation, safety, and mission progression.

III. SYSTEM ARCHITECTURE

MANAR's architecture establishes clear functional boundaries across operational layers. Fig. 1 illustrates core components and data flows.

A. Target State Ownership Architecture

MANAR's selected target architecture assigns authoritative runtime state to the C++ Control Core. Under this design, state modifications requested by operators or triggered by safety rules would be evaluated and committed exclusively by the control layer. Subsystems would report evidence and health rather than directly mutating shared state.

Current evidence boundary: several prototype subsystem functions still persist runtime data directly via `saveRuntime()`. The source therefore represents a transition toward centralized ownership, not a completed single-writer architecture.

B. Communication & IPC Architecture

The communication architecture distinguishes between current prototype IPC and the selected final interface design:

- **Current Prototype Data Flow:** File-backed JSON contracts under `runtime/` handle prototype communication and state persistence: `commands.json` serves as the command/request transport from the terminal to the control application, `runtime.json` provides the live prototype status representation, and `search_locations.json` stores the persisted locked search-plan representation written by `control.cpp` upon processing location entry commands.
- **Selected Target Architecture (planned):** Real-time WebSocket + JSON protocol between a React/TypeScript GUI and the C++ control core. In this target model, `runtime.json` is retained as an on-demand snapshot export rather than the live transport.

IV. MISSION ARCHITECTURE AND VERIFICATION POLICY

MANAR separates mission navigation from candidate verification. Sector progression, rectangular sweeps, manual rescue-state entry, and RTL state changes are represented in the C++

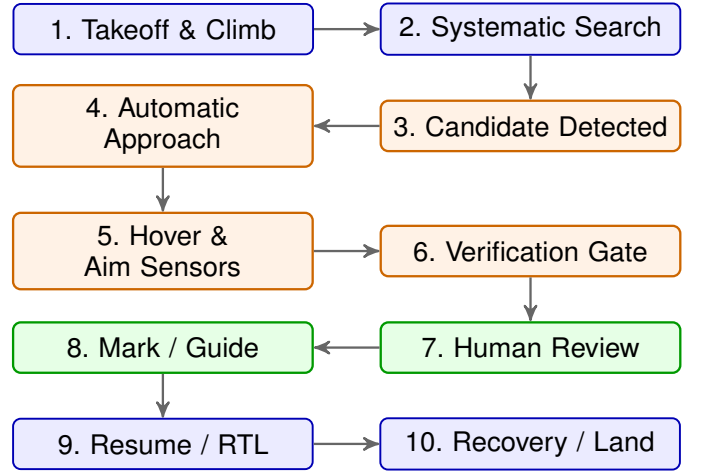


Fig. 2. Conceptual 10-stage search, verification, and rescue lifecycle.

source. The end-to-end candidate approach and synchronized multisensor sequence in this section is a target control policy and has not yet been executed by a flight controller.

A. Navigation vs. Mission Perception Layering

To protect core vehicle stability, MANAR strictly separates flight-critical navigation telemetry from mission perception and payload subsystems:

- **Navigation Sensor Interface (planned):** Standardized interfaces for GNSS, IMU, magnetometer, barometric altitude, and local clearance sensing require Pixhawk/PX4 integration.
- **Payload Perception Suite (mixed maturity):** ONNX Runtime visual inference is implemented on recorded RGB-, infrared-, and thermal-labeled clips. Calibrated physical cameras, acoustic sensing, emergency RF, and temporal fusion remain planned.

B. Conceptual Mission Lifecycle

Fig. 2 depicts the 10-stage conceptual mission lifecycle from deployment to recovery.

C. Target 9-Step Candidate Verification Policy

The selected design requires a future integrated controller to execute the following inspect-and-review policy. The sequence is presented as a testable specification, not as a verified current behavior:

- 1) **Evidence Registration:** Record candidate geodetic coordinates and optical detection confidence.
- 2) **Constraint Validation:** Validate geofence boundaries, minimum terrain clearance, and remaining battery margin.
- 3) **Velocity Deceleration:** Automatically ramp down horizontal speed to *Inspect* mode ($v_I = 2.0$ m/s).
- 4) **Altitude Transition:** Descend to designated inspection altitude H_{inspect} .
- 5) **Positioning Orbit:** Execute bounded inspection loiter if required.

- 6) **Sensor Aiming:** Orient payload cameras toward target geodetic coordinates.
- 7) **Hover Stabilization:** Transition to stationary hover ($v_H = 0.0$ m/s).
- 8) **Synchronized Inspection:** Evaluate high-resolution optical frames and acoustic listening window.
- 9) **Supervisory Alert Escalation:** Transmit a high-priority advisory; the human operator assigns the final probable-rescue classification.

D. Prototype Control Flow & RTL Semantics

In the C++ prototype, multi-location search plans execute sequentially via `mission.cpp`. When sector k completes ($1 \leq k \leq N$), the control loop advances $k \leftarrow k + 1$ and re-initializes sweep counters. Upon completion of all N sectors or upon failsafe trigger, Return-to-Launch (`activateRTL()`) is commanded, resetting navigation destination to home coordinates (`isdestinationhome = true`).

V. NAVIGATION GEOMETRY AND KINEMATIC MODELS

MANAR incorporates geodetic navigation calculations, simulated kinematic bounds, and grid coverage geometry within the C++ control core.

A. Geographic Distance (Haversine Formulation)

To evaluate spatial relationships over spherical coordinates without a local planar projection, MANAR implements the standard double-precision Haversine formula [17] in `route_optimizer.cpp` and `drone.cpp`.

Let two points on the Earth's surface be defined by latitude and longitude in radians: $P_1 = (\phi_1, \lambda_1)$ and $P_2 = (\phi_2, \lambda_2)$. The angular differences are $\Delta\phi = \phi_2 - \phi_1$ and $\Delta\lambda = \lambda_2 - \lambda_1$. The square of half the chord length between the points is:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (1)$$

The angular distance in radians is $c = 2 \operatorname{atan2}(\sqrt{a}, \sqrt{1-a})$, yielding surface distance:

$$d(P_1, P_2) = R_E \cdot c \quad (2)$$

where $R_E = 6371000.0$ m is the mean Earth radius.

B. Kinematic Presets and Estimated Time of Arrival

The simulated aircraft translates at horizontal velocities defined by four operational flight modes: `Quick` ($v_Q = 15.0$ m/s for transit), `Active` ($v_A = 8.0$ m/s for lawnmower grid sweeps), `Inspect` ($v_I = 2.0$ m/s for candidate approach), and `Hover` ($v_H = 0.0$ m/s for stationary verification).

For current simulated coordinates P_{drone} and target waypoint P_{dest} , remaining distance is $d = d(P_{\text{drone}}, P_{\text{dest}})$. For active velocity $v(m) > 0$, Estimated Time of Arrival (t_{ETA}) is calculated as $t_{\text{ETA}} = d/v(m)$.

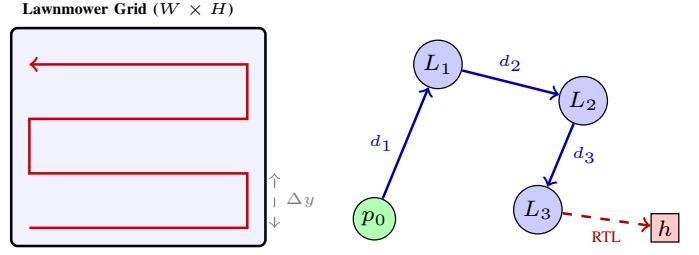


Fig. 3. Spatial geometry: Lawnmower sweep coverage pattern (left) and sequential greedy route progression with RTL return leg (right).

C. Discrete Battery Simulation Logic

To exercise safety state transitions and failsafe triggering in software without requiring physical telemetry, battery percentage $B_k \in [0, 100]$ decrements on each simulation tick by $B_{k+1} = \max(0.0, B_k - \alpha_m \Delta B)$, where $\Delta B > 0$ is a step constant and α_m is a mode multiplier ($\alpha_{\text{Quick}} = 5$, $\alpha_{\text{Active}} = 3$, $\alpha_{\text{Inspect}} = 2$, $\alpha_{\text{Hover}} = 1$). This discrete simulation rule functions strictly as a software-level unit test driver for failsafe validation, not an electro-chemical battery model.

D. Lawnmower Search Grid Geometry

For a sector centered at (ϕ_c, λ_c) with dimensions $W \times H$ (meters), local coordinate conversions use standard meridian and parallel scaling factors: $M_{\text{lat}} = 111320.0$ m/deg and $M_{\text{lon}}(\phi_c) = 111320.0 \cos(\phi_c \pi / 180.0)$ m/deg. The southwest origin corner (ϕ_0, λ_0) is computed as:

$$\phi_0 = \phi_c - \frac{H/2.0}{M_{\text{lat}}}, \quad \lambda_0 = \lambda_c - \frac{W/2.0}{M_{\text{lon}}(\phi_c)} \quad (3)$$

Parallel sweep tracks are generated along longitude separated by row spacing Δy .

VI. SEARCH ROUTE OPTIMIZATION AND HEURISTIC EVALUATION

When human operators enter multiple search sectors, the transit sequence directly impacts total flight distance and mission duration. MANAR incorporates a deterministic nearest-neighbor route planner within the C++ control core. Fig. 3 illustrates the spatial sweep geometry and waypoint routing graph.

A. Problem Formulation & Heuristic Selection

Let an operator designate an unordered set of n target search locations $L = \{L_1, L_2, \dots, L_n\}$, with $L_i = (\phi_i, \lambda_i)$. Let $p_0 = (\phi_{\text{drone}}, \lambda_{\text{drone}})$ denote initial position and $h = (\phi_{\text{home}}, \lambda_{\text{home}})$ denote home base coordinates. Finding the globally minimal tour through n arbitrary points is an instance of the metric Traveling Salesperson Problem (TSP) [8], requiring $n!$ route evaluations under brute-force search.

For embedded mission initialization where candidate sectors are modest ($n \leq 20$), MANAR selects the classical $O(n^2)$ greedy nearest-neighbor heuristic. Starting from p_0 , the planner

TABLE III
FIXED-SEED ROUTE EVALUATION: GREEDY VS. 2-OPT REFINEMENT

n	Trials	Greedy Mean (km)	2-Opt Mean (km)	Relative Excess Mean $\pm$ SD (Max)
5	100	11.42	10.92	$4.63\% \pm 5.30\%$ (22.31%)
8	100	14.75	13.69	$7.97\% \pm 7.41\%$ (29.46%)
10	100	16.15	14.90	$8.61\% \pm 8.12\%$ (31.21%)
15	100	19.50	17.96	$8.56\% \pm 7.13\%$ (26.40%)
20	100	21.81	19.63	$11.09\% \pm 7.63\%$ (32.80%)

iteratively selects the closest unvisited location under Haversine distance:

$$p_{k+1} = \operatorname{argmin}_{x \in U_k} d(p_k, x), \quad k = 0, 1, \dots, n-1 \quad (4)$$

where U_k is the unvisited set ($U_0 = L$, $U_{k+1} = U_k \setminus \{p_{k+1}\}$). The home base h is appended solely as the final Return-to-Launch return leg $d(p_n, h)$:

$$d_{\text{total}} = d(p_0, p_1) + \sum_{k=1}^{n-1} d(p_k, p_{k+1}) + d(p_n, h) \quad (5)$$

B. Fixed-Seed Reference Evaluation

The reproducibility script `reproducibility/route_benchmark.py` evaluates 500 layouts using Python's fixed pseudorandom seed 20,260,815. For each $n \in \{5, 8, 10, 15, 20\}$, 100 target sets are sampled uniformly in a 5 km $\times$ 5 km square centered at (25.2048°, 55.2708°). Start and home coincide at the square center. Each nearest-neighbor tour is refined by repeated best-improvement 2-opt edge reversal until no improving reversal remains. The committed CSV contains the unrounded aggregate values.

The reported relative excess is

$$g_{2\text{opt}} = 100 \left(\frac{d_{\text{greedy}}}{d_{2\text{opt}}} - 1 \right) \%. \quad (6)$$

This quantity is *not* an optimality gap: 2-opt is a local heuristic and can itself remain above the global optimum. Exact TSP comparison for small n is future work.

As summarized in Table III:

- 1) **Route quality:** Mean excess over the 2-opt-refined reference rises from 4.63% at $n = 5$ to 11.09% at $n = 20$. The non-monotonic $n = 10$ and $n = 15$ means show that a fixed number of random layouts should not be overinterpreted as a smooth complexity law.
- 2) **Tail behavior:** Worst observed excess ranges from 22.31% to 32.80%, so nearest-neighbor ordering can be materially worse even when its average is moderate.
- 3) **Runtime scope:** This reference script deliberately omits timing because Python runtime would not establish the C++ implementation's latency. A C++ benchmark must disclose compiler, optimization flags, processor, warm-up, repetitions, and raw samples before a microsecond claim is restored.

C. Tactical Operator Override Privilege

In practical SAR missions, human domain intelligence (e.g., wind drift, survivor travel vectors, sun angle, or terrain barriers) frequently dictates a search sequence that differs from purely geometric proximity. Crucially, MANAR formalizes this via an explicit **Operator Override Privilege** toggle (Y/N). If the operator accepts optimization (Y), the greedy sequence executes; if the operator rejects optimization (N), the manual entry order ($p_k = L_k$) is strictly preserved. In both cases, submitting the plan locks the mission configuration (`searchplanlocked = true`), preventing in-flight race conditions or unexpected waypoint mutations.

VII. VISUAL PERCEPTION PIPELINE AND DIAGNOSTIC CLIP STUDY

MANAR includes a C++ `Yolo11nDetector` wrapper and Python/C++ evaluation harnesses using `ONNX Runtime` [18] and `OpenCV`. The detector produces instantaneous person-class events; tracking, identity persistence, geolocation, and rescue confirmation are deliberately outside this layer:

$$\mathbf{y}_k = \{\mathcal{B}_i, c_i\}_{i=1}^{M_k}, \quad c_i \geq c_{\text{th}} = 0.50. \quad (7)$$

A. Tensor Contract and Export Reproducibility

Each frame $I \in \mathbb{R}^{H \times W \times 3}$ is resized with aspect-ratio-preserving letterboxing to 640×640 , converted from BGR to RGB, normalized to $[0, 1]$, and arranged as a `float32` NCHW tensor. The committed YOLO11n graph has input shape $[1, 3, 640, 640]$ and output shape $[1, 84, 8400]$. Channels 0–3 encode $[c_x, c_y, w, h]$ and channels 4–83 encode COCO class scores. Person candidates are mapped through the inverse letterbox transformation and filtered with host-side non-maximum suppression at $\text{IoU}_{\text{NMS}} = 0.45$.

The static graph was exported with the following recorded command:

```
yolo export model=yolo11n.pt format=onnx imgsz=640 \
dynamic=False opset=19
```

The artifact size is 10,720,228 bytes (10.22 MB), and its recorded SHA-256 suffix is `...00804cc5c1`. Future releases should publish the complete hash and the exact upstream checkpoint identifier in a machine-readable manifest.

B. Corpus and Protocol

The local corpus contains 26 clips under `assets/Videos/Training videos/`. Direct metadata inspection gives 240 frames per clip, 24 frames/s, and 1280×720 pixels, for 6,240 frames total. Eleven clips are designated negative by filename (No Target or False Positive), contributing 2,640 frames; 15 are designated positive (clear, moving, multiple-person, or partially concealed), contributing 3,600 frames. The filenames label RGB, infrared/low-light IR, and thermal conditions. Because the repository does not preserve capture-device, spectral-band, calibration, or transformation provenance, this paper calls the corpus *multi-condition*, not a calibrated multispectral dataset.

TABLE IV
DIAGNOSTIC 26-CLIP ZERO-SHOT COMPARISON

Recorded measure	D-FINE-N	YOLO11n
Clip means available	26	26
Session latency, mean $\pm$ SD	50.53 $\pm$ 3.15 ms	29.25 $\pm$ 0.85 ms
Reciprocal session rate	19.79 s ⁻¹	34.19 s ⁻¹
Positive clips with ≥ 1 event	6/15 (40%)	12/15 (80%)
Events on positive clips	892	1,497
Events on negative clips	0	0
Model file size	14.55 MB	10.22 MB

YOLO11n and D-FINE-N [16] are evaluated zero-shot with batch size 1, a 640×640 model input, person-class threshold 0.50, and the ONNX Runtime CPU execution provider. The committed perception/vision/benchmarks/results.md stores one mean model-session latency and total event count for each model-clip pair. It does not store bounding-box ground truth, per-frame matches, raw timing samples, warm-up policy, repeated-run order, or a machine-readable hardware manifest.

For that reason, the reported measures are strictly:

- **model-session latency:** the recorded mean time around the ONNX session call for each clip;
- **detection event:** one post-threshold person box emitted on one frame; repeated boxes on the same person count repeatedly;
- **positive-scenario coverage:** the fraction of the 15 positive-designated clips with at least one event; and
- **negative-designated events:** emitted events on the 11 clips designated as containing no person target.

These measures are not average precision, precision, recall, F1, unique-person count, or tracking accuracy.

C. Observed Results

Across the 26 recorded clip means, YOLO11n's model-session latency is 42.1% lower than D-FINE-N's, a ratio of $1.73\times$. This is not an end-to-end throughput comparison: the graph boundaries differ and YOLO's host-side NMS lies outside the session timer. Decode, resize, normalization, postprocessing, rendering, and IPC must be timed separately before an onboard frame-rate claim is made.

YOLO11n produces at least one event in twice as many positive-designated clips (12 vs. 6) and $1.68\times$ as many positive-clip events (1,497 vs. 892). Both produce zero events on the 11 negative-designated clips in this corpus. Event count is intentionally interpreted as sensitivity evidence only: it may reward duplicate detections, cannot identify missed ground-truth persons, and provides no localization score.

The result supports selecting YOLO11n as the present prototype's *diagnostic* detector because it offers wider positive-scenario coverage and lower recorded session latency under the fixed protocol. It does not establish that dense convolutional detectors are generally superior to transformer detectors. Training data, confidence calibration, graph export, target scale, and postprocessing are confounded. A publication-level comparison requires matched threshold sweeps, ground-truth boxes, at least

one additional baseline, and confidence intervals over repeated end-to-end runs.

D. Planned Perception Extensions

The next evidence milestones are (1) bounding-box annotation with target size and occlusion attributes, (2) held-out scenario/site splits, (3) threshold and input-resolution sensitivity studies, (4) slicing or tiling for small targets [11], and (5) multi-frame association using a tracker such as ByteTrack [19]. Acoustic, radar, RF, and Mamba-based temporal fusion [20] remain planned and are not evaluated in this paper.

VIII. ANALYTICAL TELEMETRY POLICY [PLANNED]

To study bandwidth–energy tradeoffs before radio integration, MANAR proposes an adaptive telemetry policy and a first-order airtime model. Neither the rate controller nor its energy savings are implemented or measured in the present artifact. The analysis is retained as a design hypothesis for future hardware trials.

A. Analytical Radio Telemetry Power Model

Radio frequency (RF) telemetry power consumption is approximated as scaling linearly with payload airtime. Let P_{TX} denote transmitter active power, P_{RX} denote receiver baseline power, and R denote link channel data rate in bits per second (bps).

For a telemetry stream i with message payload size B_i (bytes) transmitted at frequency f_i (Hz), the incremental electrical radio power ΔP_i consumed over the idle/receive baseline is formulated as:

$$\Delta P_i = (P_{TX} - P_{RX}) \cdot \left(\frac{8 \cdot B_i \cdot f_i}{R} \right) = \beta \cdot B_i \cdot f_i \quad (8)$$

where β is the radio transmission slope coefficient:

$$\beta = \frac{(P_{TX} - P_{RX}) \cdot 8}{R} \left(\frac{\text{W}}{\text{byte} \cdot \text{Hz}} \right) \quad (9)$$

For an illustrative link with $P_{TX} = 4.80$ W, $P_{RX} = 0.36$ W, and channel rate $R = 64\,000$ bit/s, the slope coefficient evaluates to $\beta \approx 5.55 \times 10^{-4}$ W/(byte · Hz). The model excludes headers, acknowledgements, contention, retransmission, state-transition energy, coding, idle policy, and rate adaptation; it must not be read as a radio power measurement.

Fig. 5 illustrates the incremental radio power across transmission frequencies from 0 to 10 Hz for fourteen distinct MANAR telemetry message categories.

B. Adaptive Transmission Rate Policy

Table V specifies a candidate dual-regime rate policy for later implementation and measurement.

Under the proposed policy, routine navigation streams update at 2 Hz, slow-varying battery state at 0.2 Hz, and discrete states on change. A candidate event would switch relevant streams to 5–10 Hz. Future work must implement the scheduler, measure packet-level traffic and energy, and test whether the increased update rate improves operator localization enough to justify its cost.

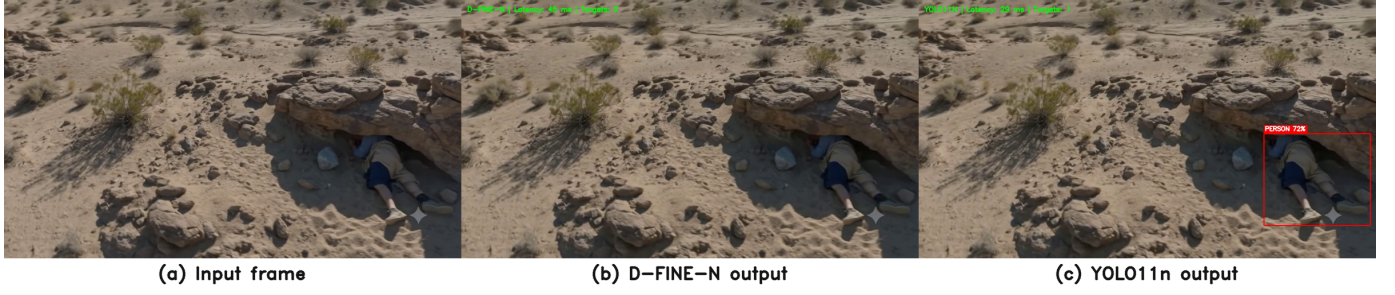


Fig. 4. Frame-aligned qualitative example from the positive-designated RGB Partially Concealed Rescuee clip at frame 239. D-FINE-N emits no person event at the fixed threshold, whereas YOLO11n emits one. This selected example illustrates the event definition; without ground-truth boxes it is not an accuracy estimate.

TABLE V
MANAR ADAPTIVE TELEMETRY RATE POLICY AND OPERATIONAL RATIONALE

Tier	Telemetry Stream	Payload (B_i)	Normal Rate	Proximity / Critical Rate	Operational Rationale
●	Heartbeat	24 B	2.0 Hz	2.0 Hz	Low-overhead link-health and watchdog awareness.
●	Position	28 B	2.0 Hz	5.0 Hz	Primary navigation telemetry; modest radio power cost.
●	Speed	24 B	2.0 Hz	5.0 Hz	Low power footprint; useful for live kinematic context.
●	Altitude	24 B	2.0 Hz	5.0 Hz	Critical for terrain clearance and vertical safety.
●	Heading	22 B	1.0 Hz	5.0 Hz	Low marginal cost; 1 Hz baseline sufficient in transit.
●	Battery Percentage	22 B	0.2 Hz	0.5 Hz	5-second interval sufficient under normal discharge.
●	Flight Mode	21 B	On change	On change	Event-driven update; redundant transmissions avoided.
●	Mission State	21 B	On change	On change	State transition telemetry published on event trigger.
●	Component Status	24 B	On change / fault	Active components only	Payload status updates sent on state modification.
●	Sensor Measurements	52 B	Local only	5.0–10.0 Hz	High-rate raw sensor data streamed only in proximity.
●	Single-Sensor Confidence	28 B	Immediate event	Stream relevant sensor	Triggered immediately upon validated detection event.
●	Multisensor Confidence	32 B	Immediate event	Trigger proximity mode	High-priority event; escalates system telemetry tier.
●	Fault / Failsafe / Low Batt	28 B	Immediate event	Repeat while active	Emergency telemetry; bypasses normal bandwidth caps.
●	Diagnostics / Logs / Config	84 B	On-demand	On-demand	High payload footprint; retrieved only on operator query.

IX. SAFETY, AUTHORIZATION, AND DETERMINISTIC FAILSAFES

The repository contains source-level command clamps, simulated battery thresholds, payload state toggles, plan locking, and RTL state transitions. The broader operational rules below are safety requirements for future integrated testing; they are not certified flight protections.

A. Supervised Operator Authorization Rules

The target system is required to enforce six supervisory rules:

- 1) **Human Confirmation Authority:** Machine-learning models may gate candidate alerts, but only the human operator may assign a probable-rescue classification.
- 2) **Multi-Sensor Consensus:** No single sensor modality dominates victim confirmation; multi-modal evidence correlation is required.
- 3) **Supporting Modality Principle:** Thermal LWIR and emergency RF detections are treated strictly as supporting cues, avoiding false alarms from heated terrain or abandoned beacon devices.
- 4) **Guidance Authorization:** Future physical signaling devices (beacon, strobe, spotlight, smoke) require explicit operator authorization before activation.
- 5) **Restricted Smoke Deployment:** Smoke marker release is strictly prohibited near fire-prone vegetation, fuel storage, crowds, or adverse wind conditions.
- 6) **Safety Precedence:** Vehicle safety limits, structural thresholds, icing risk, and minimum return energy strictly override search or detection confidence.

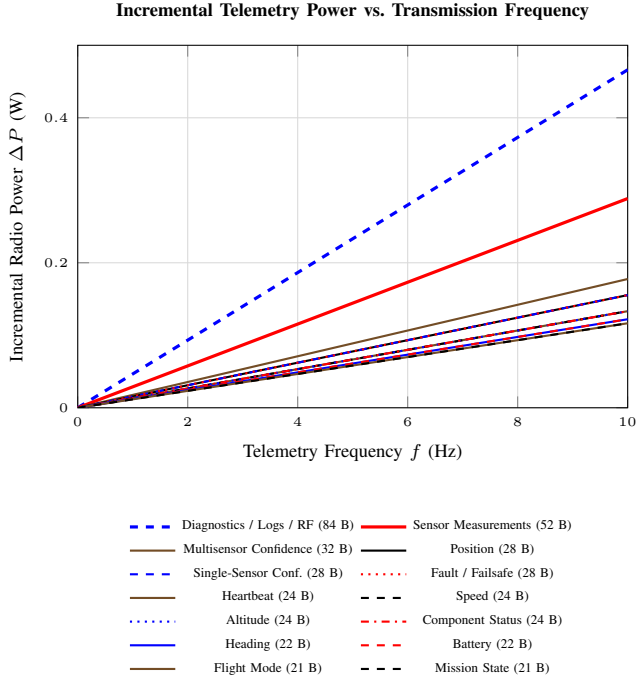


Fig. 5. Analytical estimate of incremental radio power for individual MANAR telemetry message types across transmission frequencies ($P_{TX} = 4.8$ W, $P_{RX} = 0.36$ W, $R = 64$ kbps).

B. Communications Resilience and Link-Loss Logging

The selected architecture separates low-bandwidth state telemetry from high-bandwidth video (*SYN-R-03*). Link-loss detection, onboard non-volatile logging, and autonomous return behavior require communication and autopilot integration and remain planned.

C. Battery Failsafe Hierarchy

The control core evaluates simulated battery percentage B_k on each loop iteration against a 4-tier threshold hierarchy:

- **Battery Warning** ($B_k \leq 30\%$): Emits a non-preemptive advisory warning to the ground station.
- **Recommend RTL** ($B_k \leq 25\%$): Advises the operator to terminate search sweeps and initiate return transit.
- **Emergency RTL** ($B_k \leq 15\%$): Automatically preempts active search sweeps, clears pending waypoints, and engages Return-to-Launch navigation to home base.
- **Emergency Land** ($B_k \leq 5\%$): Commands an immediate controlled vertical landing at the current coordinate position.

D. Software Command Clamps

All velocity and altitude command arguments are validated against software bounds prior to state mutation: $v_{cmd} \leq v_{max} = 15.0$ m/s and $H_{cmd} \leq H_{max} = 2000.0$ m. Commands exceeding these limits are automatically clamped, logging warning events to `runtime/logs.txt`.

X. SOFTWARE ARCHITECTURE

MANAR is implemented as a modular C++ system structured into standalone applications and decoupled domain subsystems.

TABLE VI
MANAR C++ MODULE ARCHITECTURE

Module	Type	Core Responsibilities
<code>control.cpp</code>	App	Control loop, JSON command dispatch, runtime state serialization.
<code>terminal.cpp</code>	App	Coordinate entry, DMS parsing, optimization toggle, operator commands.
<code>setup.cpp</code>	App	Configuration slot selection, parameter template initialization.
<code>mission.cpp</code>	Core	Lawnmower sweep progression, search plan locking, RTL pre-emption.
<code>drone.cpp</code>	Core	Geodetic kinematics, distance-to-home, discrete battery simulation.
<code>flight.cpp</code>	Core	Velocity/altitude clamps, flight modes (Quick, Active, Inspect, Hover).
<code>route_optimizer.cpp</code>	Core	Haversine geodetics, $O(n^2)$ greedy nearest-neighbor solver.
<code>components.cpp</code>	Core	State control for 11 controllable payload categories.
<code>shared.cpp</code>	Core	Timestamped structured plaintext event logging.
<code>yolo11n_detector.cpp</code>	CV	Reusable YOLO11n ONNX Runtime inference wrapper, letterbox preprocessing, NMS.
<code>yolo11n_video.cpp</code>	CV App	Video file evaluation harness with annotated frame output.
<code>yolo11n_live.cpp</code>	CV App	Live camera / RTSP stream detection test application.
<code>dfine_video.cpp</code>	CV App	D-FINE-N reference baseline video evaluation harness.

A. C++ Core & Perception Domain Modules

Table VI summarizes the architectural responsibilities of the core C++ source modules across control and perception subsystems.

B. Separation of Software Layers

MANAR enforces architectural separation across decoupled operational layers:

- **C++ Control Core (`core/`):** Executes deterministic mission state machine, kinematic flight bounds, and greedy route optimization.
- **Visual Perception Library (`perception/vision/`):** ONNX Runtime [18] C++ detector module and benchmark harnesses producing instantaneous candidate events.
- **Operator Interface (`planned`):** Selected React/TypeScript interface and WebSocket + JSON protocol; the current control prototype uses file-backed JSON exchange.
- **Temporal Fusion Engine (`Mamba SSM`):** Planned multi-modal evidence association and candidate alert escalation filter [20].

XI. ARTIFACT AND REPRODUCIBILITY EVIDENCE

The public repository is the evidence boundary for this pre-flight paper. Table VII maps the central claims to present artifacts and names the evidence still missing. This replaces earlier percentage pass-rate claims for which no matching automated test suite or raw logs were checked in.

TABLE VII
CLAIM-TO-ARTIFACT EVIDENCE MAP

Claim area	Present artifact and supported statement	Evidence not yet present
Route ordering	<code>core/system/route_optimizer.cpp</code> implements Haversine distance and nearest-neighbor ordering with a return leg.	C++ timing harness, exact-optimum comparison, and automated edge-case tests.
Mission state	<code>mission.cpp</code> contains sector progression, plan-lock state, manual rescue-state input, and RTL state changes.	Machine-readable expected/observed traces, fault injection, SITL/HITL, and flight execution.
Command and battery safeguards	Source contains configuration clamps and a discrete battery variable used to trigger state changes.	Unit tests, energy model calibration, autopilot enforcement, and physical battery telemetry.
Visual inference	<code>perception/vision/</code> contains ONNX models, harnesses, source, and <code>benchmarks/results.md</code> .	Ground-truth annotations, environment manifest, raw timing samples, threshold sweeps, and independent test split.
Route reference study	<code>docs/paper/reproducibility/route_benchmark.py</code> and its CSV reproduce Table III with a fixed seed.	Exact TSP baseline for small n and compiled C++ runtime measurements.
Recorded clip corpus	26 local MP4 files provide 6,240 frames at 1280×720 and 24 frames/s.	Public license/provenance, capture metadata, calibration, subject documentation, and bounding boxes.
Physical system	Architecture and requirements specify selected interfaces and safety policies.	Integrated airframe, calibrated payload, legal/operational approvals, and controlled flight trials.

TABLE VIII
PROTOTYPE CONFIGURATION VALUES

Symbol	Parameter	Value
v_Q	Quick velocity	15.0 m/s
v_A	Active velocity	8.0 m/s
v_I	Inspect velocity	2.0 m/s
v_H	Hover velocity	0.0 m/s
H_{launch}	Launch altitude	10.0 m
H_{max}	Configured altitude clamp	2000 m
v_{max}	Configured speed clamp	15.0 m/s
$W \times H$	Rectangular sector	100×100 m
Δy	Sweep spacing	20.0 m
R_{reach}	Waypoint radius	5.0 m
R_{home}	Home radius	100 m
B_{warn}	Warning threshold	30%
$B_{\text{rtl_w}}$	Recommend RTL	25%
B_{emg}	Emergency RTL	15%
B_{land}	Emergency land	5%

A. Reproduction Entry Points

The route table is reproduced with:

```
python docs/paper/reproducibility/route_benchmark.py
```

The diagnostic vision pipeline is implemented by `perception/vision/benchmarks/batch_benchmark.py`; it requires the local clip corpus, the two committed ONNX graphs, OpenCV, NumPy, and ONNX Runtime. A publication artifact should add a lockfile, complete model hashes, a hardware/OS manifest, raw per-frame timing output, and a non-interactive command that regenerates every table and figure.

B. Simulated Configuration

Table VIII lists parameters used by the software prototype. They exercise state and geometry logic and do not constitute validated aircraft limits.

XII. THREATS TO VALIDITY AND FUTURE WORK

A. Construct Validity

The vision study measures emitted events, not correct detections. One visible person may generate events on many

adjacent frames, and a clip with one event counts as covered regardless of localization. Without boxes and an IoU matching rule, true positives, false negatives, precision, recall, AP, F1, and unique-person counts are undefined. Likewise, the route study compares greedy construction with a 2-opt local optimum; its relative excess is not a global optimality gap.

B. Internal Validity

The committed detector record contains one aggregate latency per model-clip pair, not raw repeated measurements. Warm-up, processor state, thread settings, run order, and clock variability are not preserved. The session graphs also have asymmetric boundaries: YOLO host-side NMS is outside the timer, while the D-FINE export can include different graph-side operations. The 0.50 confidence threshold is not calibrated per model. Therefore the observed $1.73\times$ latency ratio and event-count difference are diagnostic, not causal evidence about detector families.

The fixed-seed route script is reproducible but samples one square geometry and one start/home rule. Different spatial distributions, separated start and home, priority weights, obstacles, wind, and energy costs may change the ranking. The Python study also does not establish C++ execution time.

C. External Validity and Data Provenance

The 26 clips are short, locally stored, temporally autocorrelated, and not divided into development and held-out test sites. Their filenames describe RGB, infrared, and thermal conditions, but the repository does not document camera models, spectral bands, calibration, altitude, date, location protocol, postprocessing, subject consent, or data license. Results cannot be generalized to physical LWIR payloads, other terrain, weather, altitudes, cameras, or populations until this provenance and independent testing exist.

D. Software and Operational Validity

Source presence is not equivalent to verified behavior. The repository does not yet include an automated suite reproducing

the previously claimed route, plan-lock, clamp, IPC, memory, and RTL pass rates. No claim is made for memory-leak freedom, real-time scheduling, communication loss recovery, collision avoidance, geofencing, airworthiness, regulatory compliance, or all-weather operation. The analytical battery and telemetry models are test/design aids rather than physical models.

Environmental hazards remain substantial: precipitation and obscurants degrade optics; thermal crossover and sun-heated terrain create clutter; wind and rotor vibration affect acoustics and control; terrain and structures create RF multipath; and icing or dust can threaten propulsion. None has been validated in flight.

E. Priority Evidence Milestones

The next work should proceed in evidentiary order:

- 1) publish a data manifest, lawful provenance, and bounding-box annotations with target size and occlusion attributes;
- 2) create development/site-held-out splits and evaluate AP, recall, false positives per image, and confidence intervals;
- 3) add matched detector baselines, threshold/resolution/tiling ablations, and frame-aligned failure cases;
- 4) benchmark end-to-end latency and power on the intended onboard computer with raw repeated samples;
- 5) add exact TSP comparisons for small n and a compiled C++ timing harness;
- 6) check in unit/integration tests and expected/observed state traces for plan locking, clamps, malformed IPC, battery transitions, and RTL precedence;
- 7) progress through PX4 SITL, HITL, tethered tests, controlled outdoor flight, and only then operationally representative trials; and
- 8) evaluate operator workload, alert review time, missed candidates, and false dispatches before claiming human-supervision benefit.

XIII. CONCLUSION

This paper recast MANAR as an evidence-scoped pre-flight software testbed rather than a flight-ready rescue system. The current artifact represents Haversine navigation, nearest-neighbor sector ordering, manual-order preservation, plan locking, RTL state changes, and ONNX detector harnesses in source. A fixed-seed 500-instance route study shows that nearest-neighbor tours exceed a 2-opt-refined local-search reference by 4.63%–11.09% on average across the evaluated target counts. A separate 26-clip diagnostic records lower model-session latency and wider positive-scenario coverage for YOLO11n than D-FINE-N, but the absence of bounding-box ground truth restricts interpretation to raw events and clip coverage. The most important result is therefore the boundary itself: the repository supports a reproducible software baseline and several design hypotheses, not physical endurance, detector accuracy, multisensor fusion, or operational rescue performance. Closing

that gap requires annotations, broader baselines, automated tests, end-to-end target-hardware measurements, SITL/HITL, and controlled flight evidence.

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